NPTEL Online Certification Courses

Indian Institute of Technology Kharagpur
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Course Name: Blockchain and its Applications (NOC26_CS34)

Assignment 4 - Week 4 (Jan 2026)

TYPE OF QUESTION: MCQ/MSQ

Number of questions: 10 **Total mark: 10 X 1 = 10**

QUESTION 1

Which of the following scenarios describes a "Soft Fork" in the Bitcoin network?

a) A protocol change where the new rules are stricter, making old blocks valid but new blocks potentially invalid to non-upgraded nodes.
b) A divergence where the blockchain splits permanently into two incompatible chains (e.g., Bitcoin and Bitcoin Cash).
c) A backward-compatible upgrade where non-upgraded nodes can still validate new blocks.
d) A temporary split caused by two miners finding a block simultaneously.

Answer: (c)

Detailed solution:
A soft fork is a backward-compatible upgrade. Old nodes will still recognize the new blocks as valid, unlike a hard fork which requires all nodes to upgrade.

QUESTION 2

Suppose a miner initially receives 100 bitcoins as a reward for successfully mining a block at time Jan, 2009. The reward for mining a block is halved approximately every four years (or after every 210,000 blocks). Based on this halving process, which of the following statements are correct? (Please note that once the reward is halved, it will remain the same until four years have been completed or after every 210,000 blocks.)

a) In Jan 2013, the miner will receive 50 bitcoins for adding a new block.
b) In Jan 2018, the miner will receive 25 bitcoins for adding a new block.
c) In Jan 2021, the miner will receive 12.5 bitcoins for adding a new block.
d) In Jan 2024, the miner will receive 6.25 bitcoins for adding a new block.

Answer: (a), (b), and (c)

Detailed solution:
The Bitcoin block reward halves approximately every 4 years (after 210,000 blocks). From Jan 2009 reward was 100 bitcoins(same for 2010, 2011, and 2012); By Jan 2013, the reward was 50 bitcoins(same for 2014, 2015, and 2016); by Jan 2018 it was 25 bitcoins(same for 2017, 2019, and 2020); and by Jan 2021 it was 12.5 bitcoins(same for 2022, 2023, and 2024). By Jan 2024, the halving to 6.25 bitcoins will not yet have occurred, making the last statement incorrect.

QUESTION 3

In the absence of a central authority, why is the "Longest Chain Rule" crucial for preventing double-spending?

a) It ensures that the chain with the most cumulative computational work is accepted as the single source of truth.
b) It allows users to spend the same UTXO multiple times if the chain is long enough.
c) It encrypts the transaction data so it cannot be copied.
d) It ensures that the timestamp of every block is exactly 10 minutes apart.A centralized authority will be used to verify each transaction before it is added to the blockchain.

Answer: (a)

Detailed solution:
By accepting the longest chain (most accumulated PoW), the network resolves conflicting histories (double spends), ensuring only one version of the transaction history is treated as valid.

QUESTION 4

In a permissionless blockchain like Bitcoin, which specific attack vector is mitigated by requiring a costly resource (like computing power in PoW) to participate in consensus?

a) SQL Injection
b) Sybil Attack
c) Man-in-the-Middle Attack
d) PhishingEnsuring that all participants trust a central authority

Answer: (b)

Detailed solution:
In a permissionless system, anyone can create an identity. Without a cost (PoW), an attacker could spawn millions of fake nodes (Sybil attack) to overwhelm the network.

QUESTION 5

Which component in the block header serves as a "fingerprint" for all transactions included in that specific block?

a) The Nonce
b) The Previous Block Hash
c) The Merkle Root
d) The Miner's Public Key

Answer: (c)

Detailed solution:
The Merkle Root is a hash generated from all transaction hashes in the block. If a single byte in any transaction changes, the Merkle Root changes completely.

QUESTION 6

Traditional distributed databases handle "Crash Faults" effectively. What specific, more complex class of faults is the Bitcoin blockchain designed to tolerate?

a) Hardware Failures
b) Network Latency
c) Byzantine Faults
d) Power Outages

Answer: (c)

Detailed solution:
Byzantine faults involve malicious actors who may lie or send conflicting information to different parts of the system (unlike a crash, where a node simply stops working).

QUESTION 7

Which of the following best describes **Safety** and **Liveness** in Bitcoin?

a) **Safety** ensures transactions are irreversible, while **Liveness** ensures transactions are eventually added.
b) **Safety** guarantees quick transaction confirmation, while **Liveness** prevents forks.
c) **Safety** prevents double-spending, while **Liveness** speeds up block creation.
d) **Safety** ensures blocks are always valid, while **Liveness** ensures no transaction delays.

Answer: (a)

Detailed solution:
Safety ensures that once a transaction is confirmed, it cannot be reversed, preventing issues like double-spending. **Liveness** ensures that transactions will eventually be added to the blockchain, even if there are delays or forks.
Please refer to the Week 4 Lecture 19.

QUESTION 8

Mathematically, what is a miner attempting to do during **Proof of Work** (PoW)?

a) Decrypt the previous block's signature.
b) Find a nonce such that □□□h(□□□□□□□□□□□□□) < □□□□□□.
c) Sort all transactions in the mempool alphabetically.
d) Calculate the private key of the Satoshi Nakamoto wallet.

Answer: (b)

Detailed solution:
Miners vary the nonce to produce a block hash that is numerically lower than a specific □□□□□□ value set by the network difficulty.

QUESTION 9

Before a miner begins the heavy computational work (hashing) to mine a new block, what must they typically do first?

a) Broadcast their solution to the network.
b) Construct a candidate block by selecting valid transactions from their memory pool.
c) Wait for the next difficulty adjustment period.
d) Shut down their node to save power.

Answer: (b)

Detailed solution:
The miner must first assemble the "Candidate Block" (select transactions, verify them, create Merkle root) before they can start hashing the header to solve the PoW puzzle.

QUESTION 10

What is the theoretical consequence if a single mining pool obtains >50% of the network's hashrate (The 51% Attack)?

a) They can steal funds from any user's wallet (e.g., transfer your BTC to them).
b) They can change the maximum supply of Bitcoin to 42 million.
c) They can reverse their own recent transactions (double spending) and block other transactions from confirming.
d) They can shut down the internet connection of other miners.

Answer: (c)

Detailed solution:
A 51% attacker cannot break cryptography (steal funds) or change consensus rules (supply cap), but they can rewrite recent chain history to double-spend and censor new transactions.